



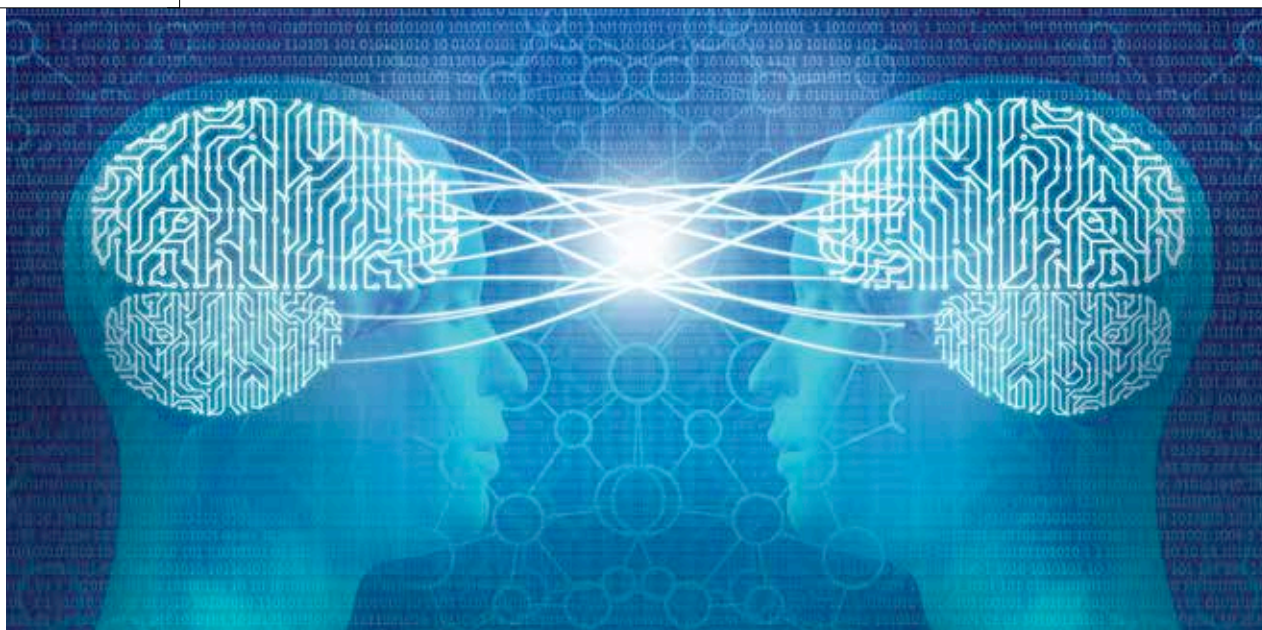
Neurotechnology of music

Imagine algorithms that can detect the onset of depression, and play bespoke music to make you happy <https://digit.in/aug19-85>



Hail the Ommissiah!

With the advent of brain computer interfaces, will the human brain begin to communicate in binary? <https://digit.in/aug19-86>



INCHING TOWARDS TELEPATHY

BrainNet, a multi-person brain-to-brain interface is bringing us closer to real-life telepathy

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hen you think about telepathy as a concept, it is likely to be in a science-fiction setting. However, in

the past few years, physicists and neuroscientists have developed an arsenal of tools that can sense certain types of thoughts and transmit information to other brains. We are approaching the possibility of brain-to-brain communication or telepathy becoming a reality. A group of neural engineers from the University of Washington have created a

system that allows three people to collaboratively play a Tetris-like game by transmitting the 'Sender's' thoughts to the 'Receiver'.

MAKING THE CONNECTION

The theory of connecting humans minds was put to test in 2015 by Andrea Stocco from the University of Washington. Two people were connected to a prototype transmitter and receiver. The device placed on the test subjects' heads similar to a shower caps, with a series of electrodes on the inside. The team achieved mostly positive results after the participants of the study played a game somewhat similar to 20-questions. BrainNet essentially builds on this contraption. A team led by Dr Rajesh Rao, a professor at the University of Washington, built BrainNet which borrows from Stocco's work as well as Rao's work back in 2014.

THE TECH BEHIND BRAINNET

The system, as seen in Stocco's work, seemed to work with two people, so

Rao went a step further and decided to incorporate yet another person in the process. In the new study, two people were dubbed 'Senders' and one person dubbed the 'Receiver' were made to play a game similar to Tetris. The senders can see both the block and the line but can't control the game, while the receiver can only see the block but is the one that actually controls the game. Basically, each sender needs to decide whether they need to flip a block to complete a line (just like Tetris). The senders were hooked up to the EEG machine and by using real-time EEG analysis, the decisions of whether or not to move a piece were decoded and then sent to Receiver over the internet. Then, transcranial magnetic stimulation (TMS) was used to activate the corresponding areas in the receiver's brains. The message received by the receiver is translated and signals are then sent to the occipital cortex of the game. When the information has been processed by the receiver, he sends a command of